

Preclass Learning: Adv Data Wrangling

P2 (ICA 7)

1. Relational Data Concepts

- **Relational Data:** Data stored across multiple tables that are linked together by shared key variables. This avoids storing redundant information (e.g., separating customer information from their order history).
- **Key Variables:**
 - **Primary Key:** A column (or set of columns) that uniquely identifies each row in a table.
 - **Foreign Key:** A column in one table that refers to the Primary Key in another table, forming the link.

2. Mutating Joins

Mutating joins add new variables from one table (B) to another table (A) by matching key values.

Join Type	Description	Key Result
inner_join	Keeps rows only if they have matching keys in both tables.	Returns the intersection of A and B.
left_join	Keeps all rows in the left table (A) and adds matching rows from the right table (B). Non-matches get NA values.	Most common join for adding attributes.
right_join	Keeps all rows in the right table (B) and adds matching rows from the left table (A).	Same as left join, but roles are reversed.
full_join	Keeps all rows from both tables (A and B).	Returns the union of A and B.

3. Filtering Joins

Filtering joins do not add variables; they only affect the rows of the first table (A) based on matches in the second table (B).

- **semi_join**: Keeps all rows in A that have at least one match in B. (Returns rows of A).
- **anti_join**: Keeps all rows in A that **do not** have a match in B. (Used for diagnosing missing matches or integrity issues).